

# Andrew Deal

## Software Engineer and Technical Generalist

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Software engineer and technical generalist building systems that have to survive users, machines, incomplete data, and operational constraints. Experience spans full-stack product engineering, identity at consumer scale, autonomous machinery, early-stage hardtech, and independently built AI products. Current focus: physical-AI deployment, evaluation, data operations, observability, and operator workflows.

**Languages and frameworks:** TypeScript, JavaScript, Python, React, Node.js, FastAPI, SQL

**Systems:** AWS, Docker, APIs, data pipelines, CI/CD, experimentation, observability

**AI and product:** LLM orchestration, model evaluation, UX research, product strategy, operator workflows

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## EXPERIENCE

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### Independent Engineer

**Physical AI · Self-employed** | JUNE 2026 TO PRESENT · SAN FRANCISCO BAY AREA

- Built an independent physical-AI lab spanning robot-policy training, fixed-seed evaluation, hardware bring-up, teleoperation, data collection, deployment, and failure analysis.
  - Trained a **52-million-parameter ACT policy** on 206 PushT episodes and 25,650 frames; built an inspectable rollout review with videos, metrics, reproducible checkpoints, and explicit failure analysis.
  - Brought up a SO-ARM101 leader and follower system through stable device routing, camera integration, and read-only communication with all six follower servos.
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### Baseline Health

**Founder and Product Engineer** | FEBRUARY 2026 TO MAY 2026

- Built an AI-native health-intelligence system combining wearable data, laboratory results, and daily behavior into longitudinal scoring and coaching.
  - Shipped the full product across React, TypeScript, Python, FastAPI, data ingestion, LLM orchestration, and deterministic audit paths; onboarded founding members and refined the loop with real longitudinal data.
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### Selected Hardtech

**Pixel Foundry · OmenAI · TurbineOne** | JANUARY 2024 TO JANUARY 2026

- Advised and led early-stage work across defense, construction, mining, AgTech, sensor-driven predictive maintenance, and edge ML in contested environments.
  - Translated operator needs, sensor data, and operational constraints into product strategy, diagnostic surfaces, model priorities, and customer workflows.
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### Bear Flag Robotics

**Product and Technical Lead · Senior Software Engineer**

**Acquired by John Deere** | FEBRUARY 2020 TO MARCH 2023

- Co-led the web platform supporting **more than 80,000 autonomously tilled acres** and **more than \$1 million in gross revenue** across major enterprise farms.
  - Designed full-stack systems and product workflows for autonomous field operations, customer visibility, and day and night deployment windows.
  - Synthesized customer evidence and coordinated autonomy, tractor engineering, and field operations across Bear Flag and John Deere.
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### Over

**Senior Software Engineer · Acquired by GoDaddy** | MAY 2019 TO OCTOBER 2019

- Led web delivery for Over Teams, helped bring the enterprise product to parity with iOS, contributed to threefold enterprise-account growth, and established frontend architecture and engineering standards.
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### Twitch

**Software Engineer, Identity** | MAY 2016 TO OCTOBER 2017

- Built OpenID Connect federation, username changes, registration experiments, and authentication-latency improvements in multi-team rollouts affecting **more than 10 million daily active users**.
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### The RSR Company

**Technical Program Lead** | JUNE 2013 TO JUNE 2014

- Led a global intranet rollout for Millennium Challenge Corporation and expansion of a Peace Corps partnership.
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#### George Mason University

BA, Philosophy, 2012

#### University of Oxford

Distinguished Exchange Student Program, 2010–2011  
Modern Philosophy, Logic, and Logic Systems

#### Reforge

Product Strategy and Product-Led Growth, 2022